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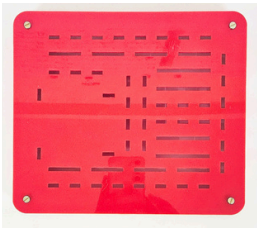
SESSION NAME

Session

7

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1



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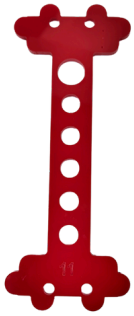


1

Spacers

2

L Clamp

3

Spacer Screws

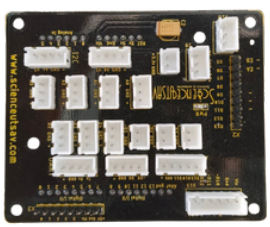
4

Nuts

5

Screw

6

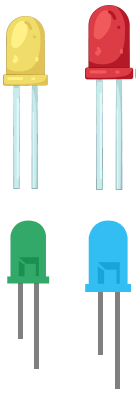


Arduino With Shield

7

Arduino cable

ELECTRONIC COMPONENTS

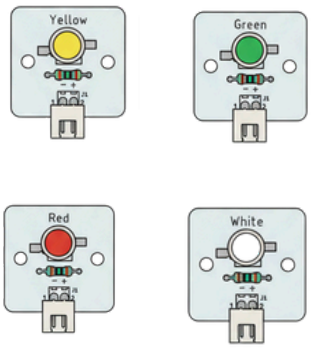


DIP LED (Dual In-line Package LED)

A DIP LED is a traditional through-hole light-emitting diode designed with two leads for easy mounting on printed circuit boards. It provides reliable and efficient illumination for indicators, displays, decorative lighting, and electronic projects, and is available in various colors, sizes, and brightness levels.

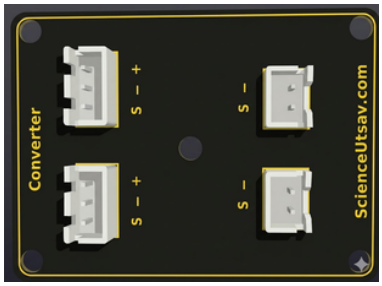
Touch Sensor

A touch sensor is an electronic input device that detects touch or proximity from a finger without requiring mechanical pressure. It provides a simple and reliable way to control electronic circuits, making it ideal for switches, automation systems, consumer electronics, and interactive projects.



1 Watt LED PCB

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



Converter

A converter is an electronic module that changes electrical power from one voltage level, current type, or signal format to another. It enables compatibility between different devices and power sources, making it widely used in power supplies, battery-operated systems, industrial equipment, and electronic projects.

String LED

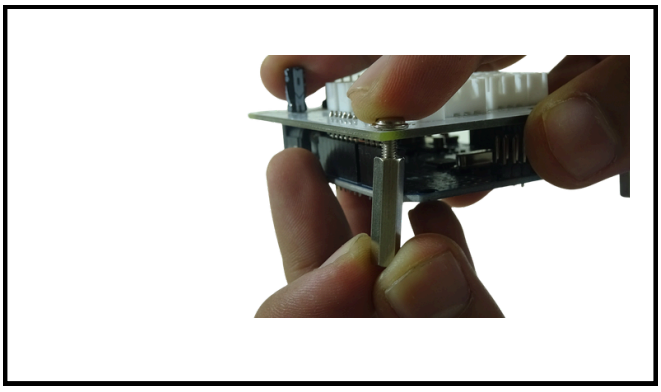
A String LED is a series of multiple LEDs connected along a flexible wire or cable to provide decorative and ambient lighting. It offers energy-efficient illumination and is widely used for home décor, festive decorations, events, signage, and creative lighting projects.

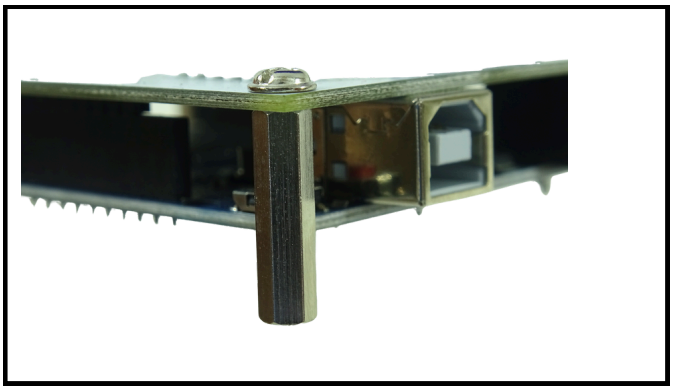
2 PIN RMC

Used to connect the 1-watt LEDs and the buzzer.

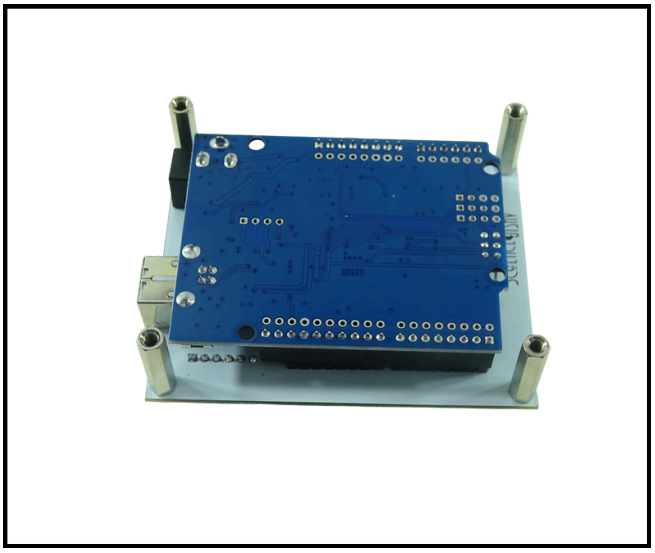
Procedure :

Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw

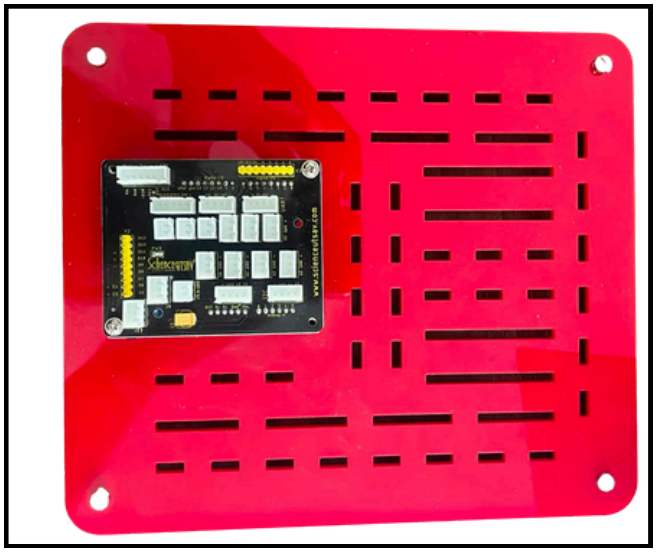




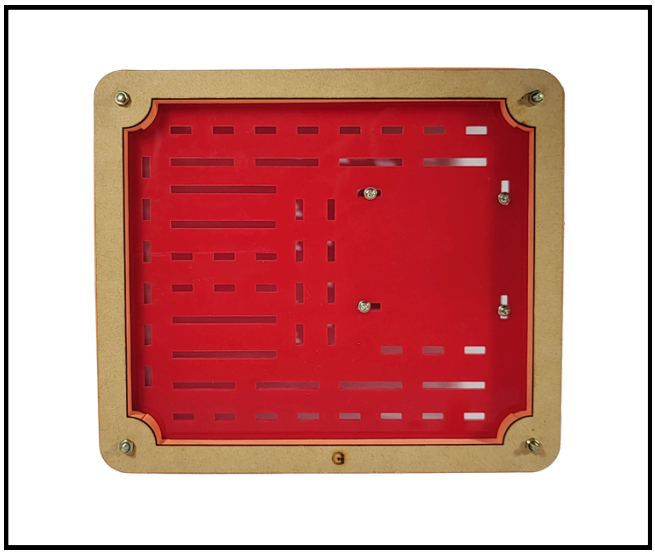
Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

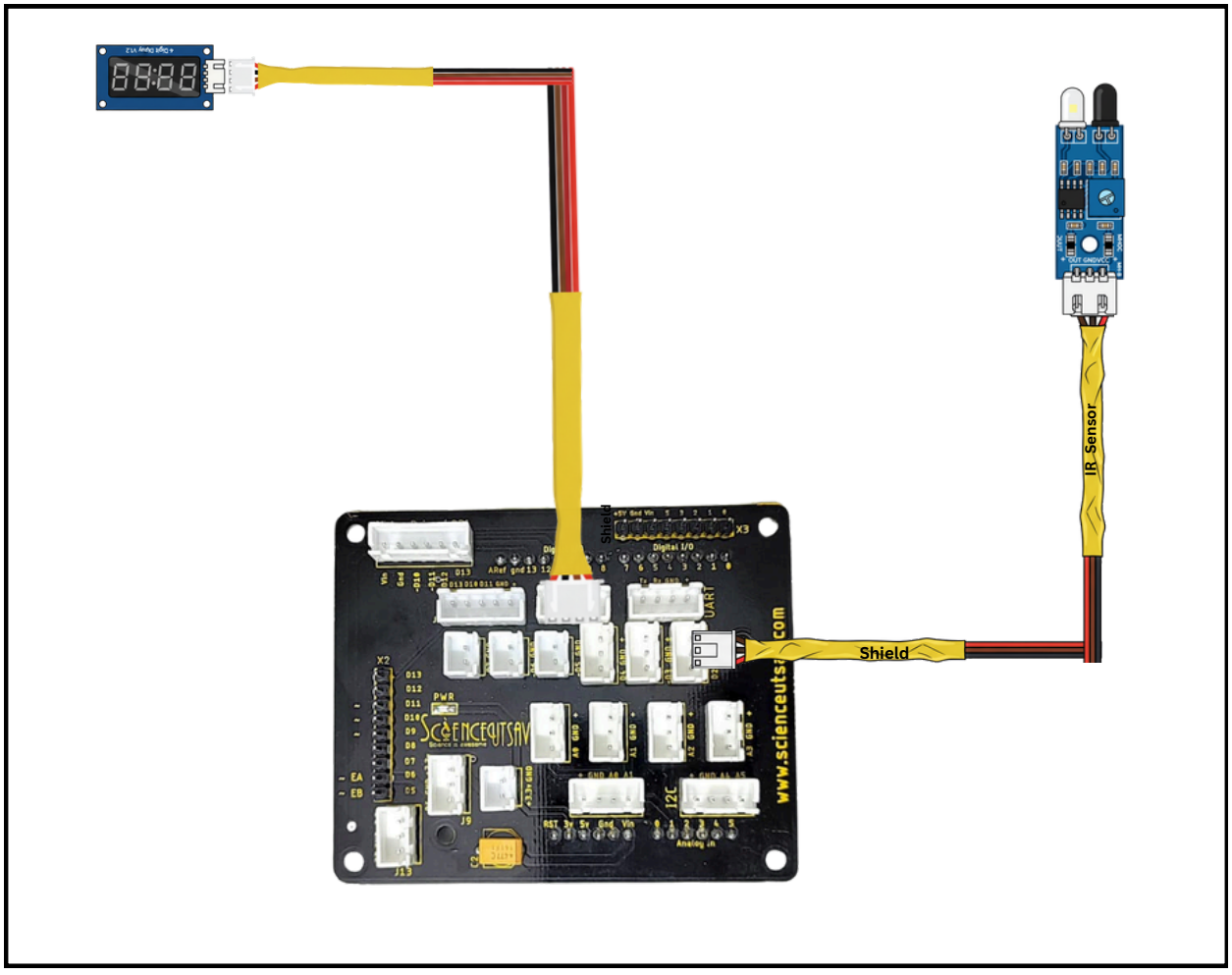


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

The circuit diagram of the project is given below

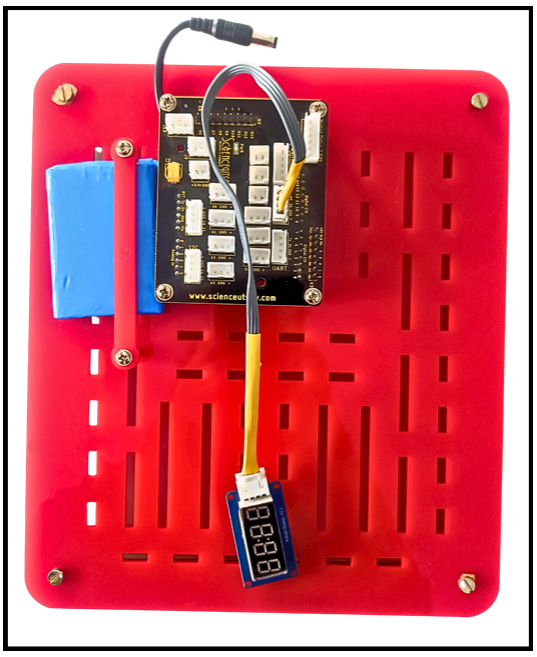


CONNECTION:

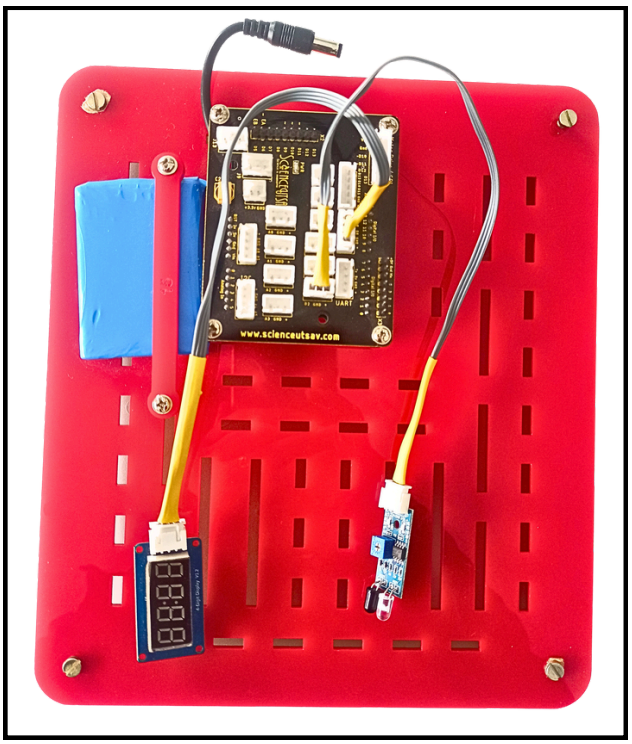
1.7 SEGMENT DISPLAY - D9 D8

2.IR SENSOR- D2

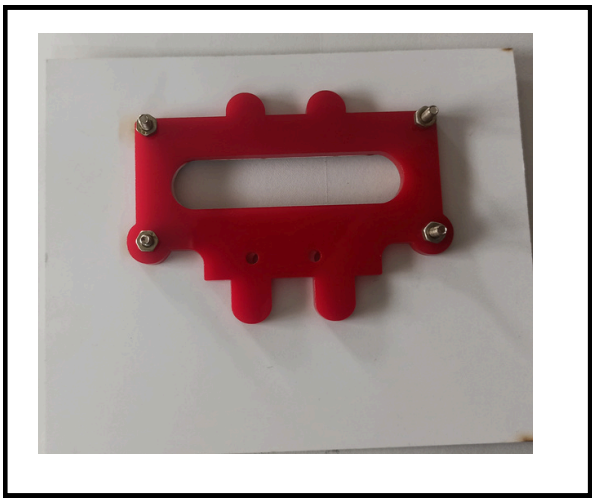
Step 1- Attach the 7 segment display to the aurdino as shown



Step 2- Connect the IR sensor as well

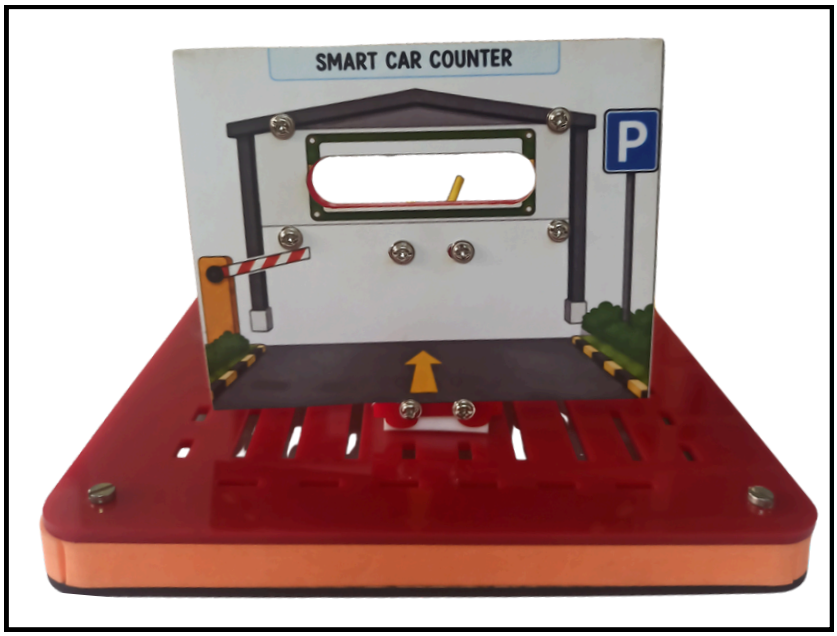


Step 3- Screw in the display stand to the template as shown



Step 4- Attach the stand shown below to the previous step

Step 5- Attach the template with the stand to the base with the help on an L-clamp



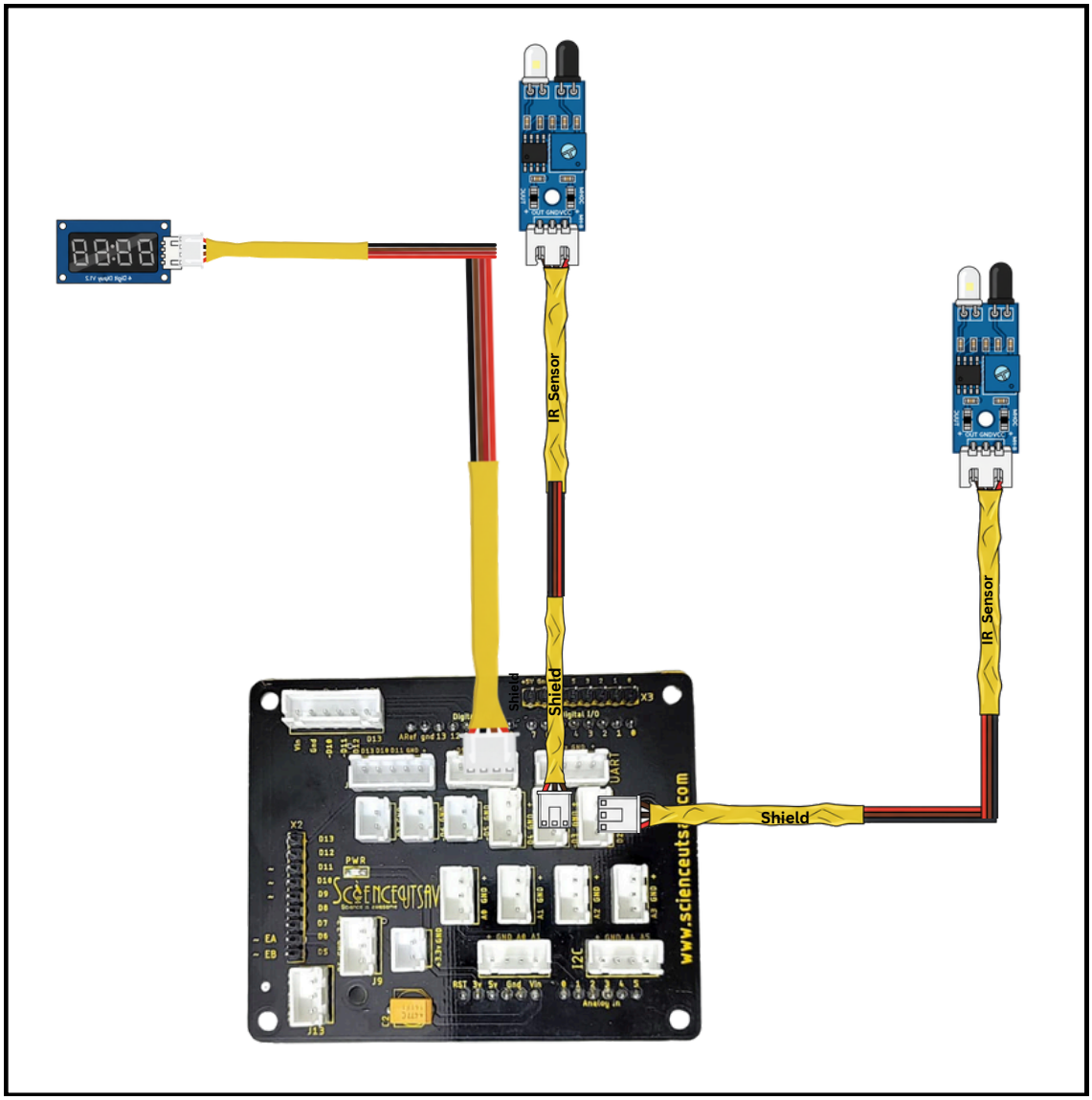
Step 6- Place the sensor and display as shown

Code

```
// Session 7 Grade 6 Project 1
 // Smart Car Counter
 // Each car passing IR sensor = count increases

 #include <TM1637Display.h>

 const int irSensor = 2;
 const int CLK = 9;
 const int DIO = 8;
 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 int carCount = 0;
 bool lastIR = HIGH;

 void setup() {
 pinMode(irSensor, INPUT);
 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 bool currentIR = digitalRead(irSensor);

 if (currentIR == IR_DETECTED && lastIR != IR_DETECTED) {
   carCount++;
   if (carCount > 9999) carCount = 0;
   display.showNumberDec(carCount, true);
   delay(300);
 }

 lastIR = currentIR;
 }
```

BEGINNER PROJECT

The circuit diagram of the project is given below



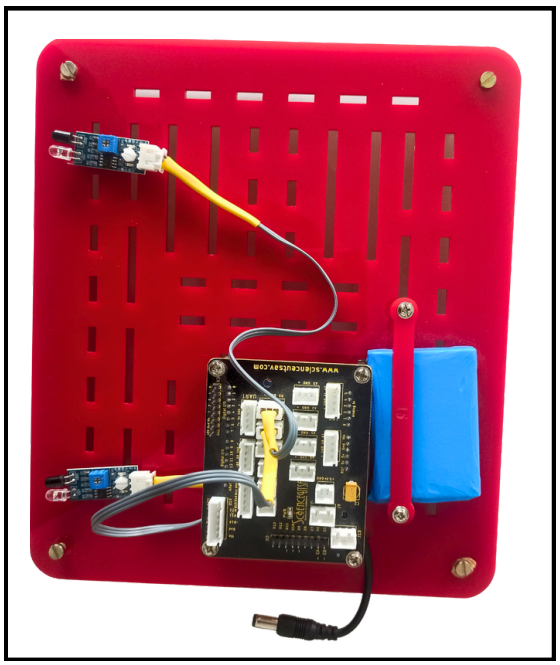
CONNECTION:

1.7 SEGMENT DISPLAY - D9 D8

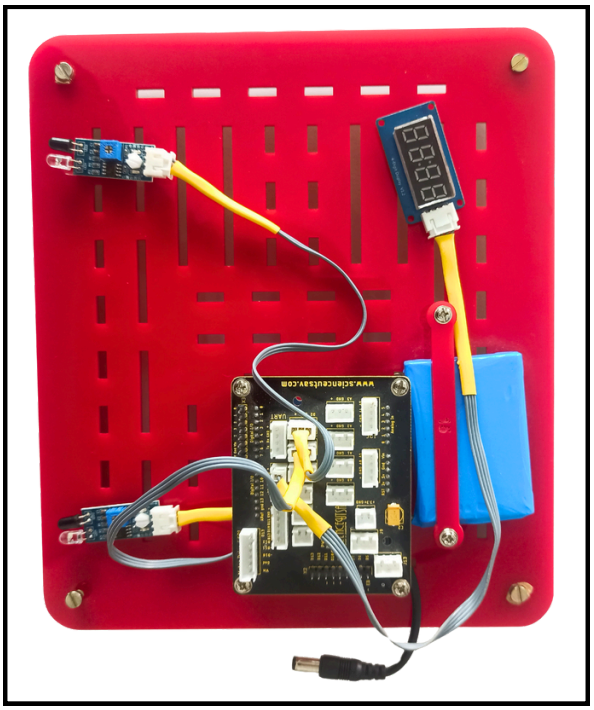
2.IR SENSOR- D2

3.IR SENSOR - D3

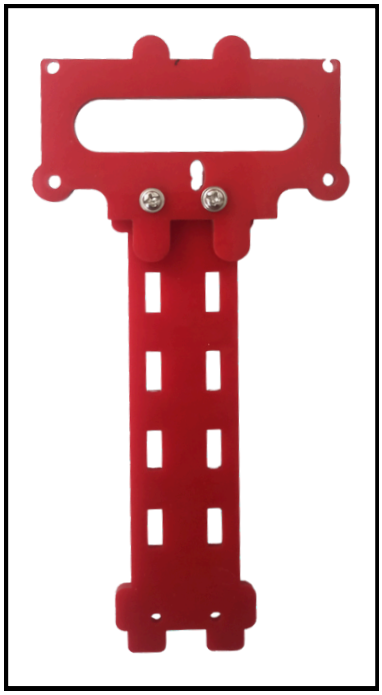
Step 1- Attach 2 IR sensors to the aurdino as shown



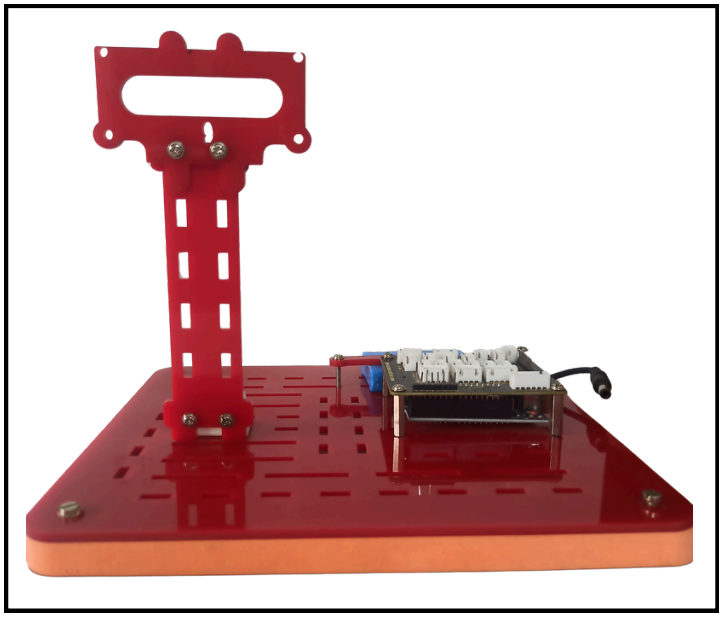
Step 2- Connect the 7 segment display to the aurdino



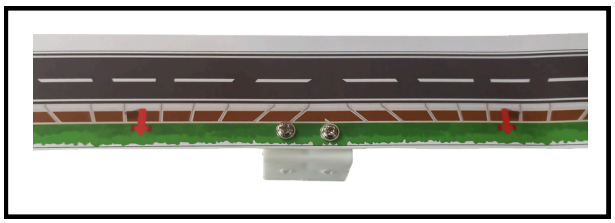
Step 3- Attach the display stand to the larger stand and secure it with bolts



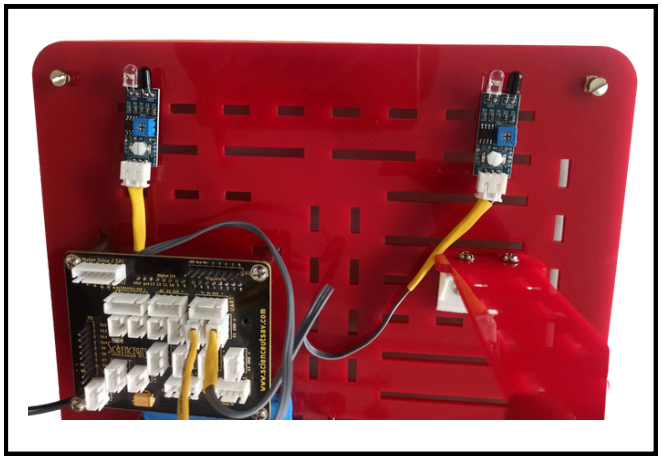
Step 4- Fix the previous stand to the base with the help of an L-clamp



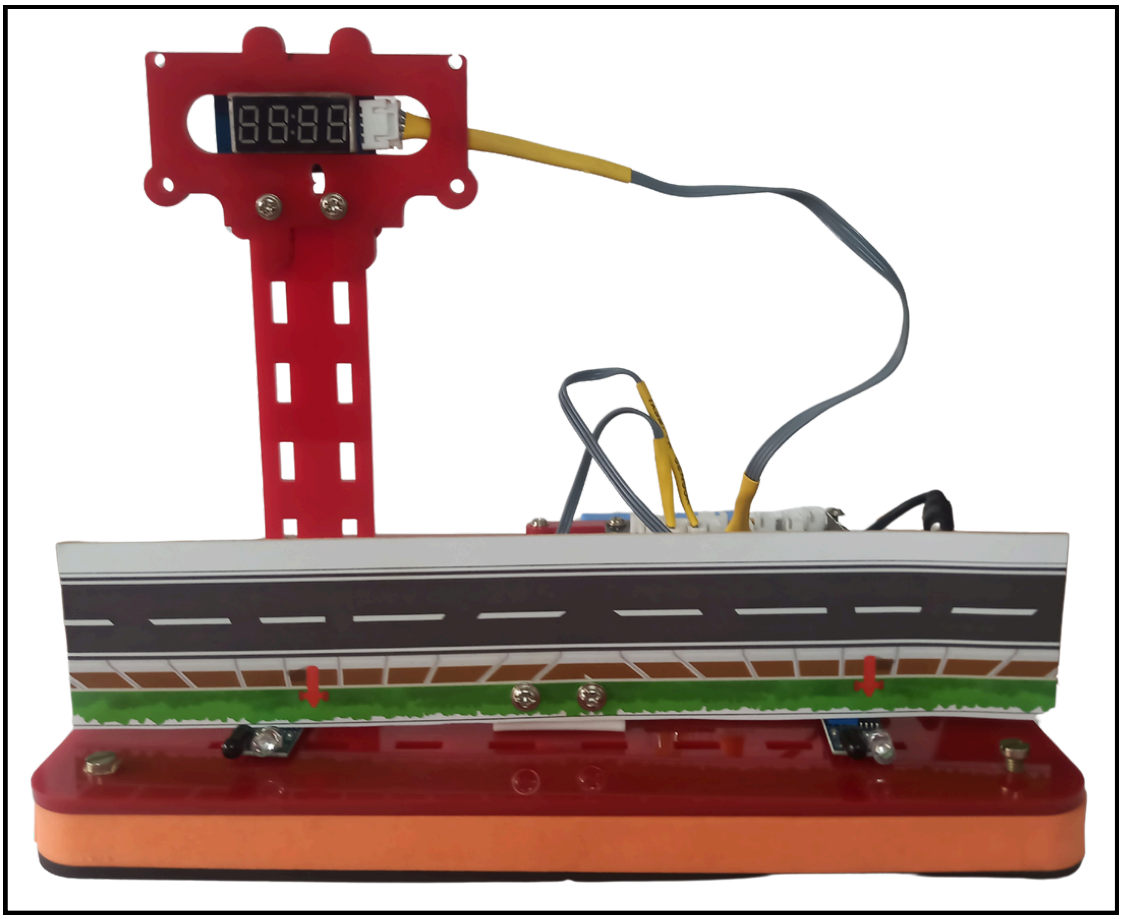
Step 5- Attach the template to L-clamp as shown



Step 6- Place and screw the IR sensors as shown



Step 7- Attach the template to the base with a L-clamp and secure it with bolts, place the display in it's stand, run the code and observe



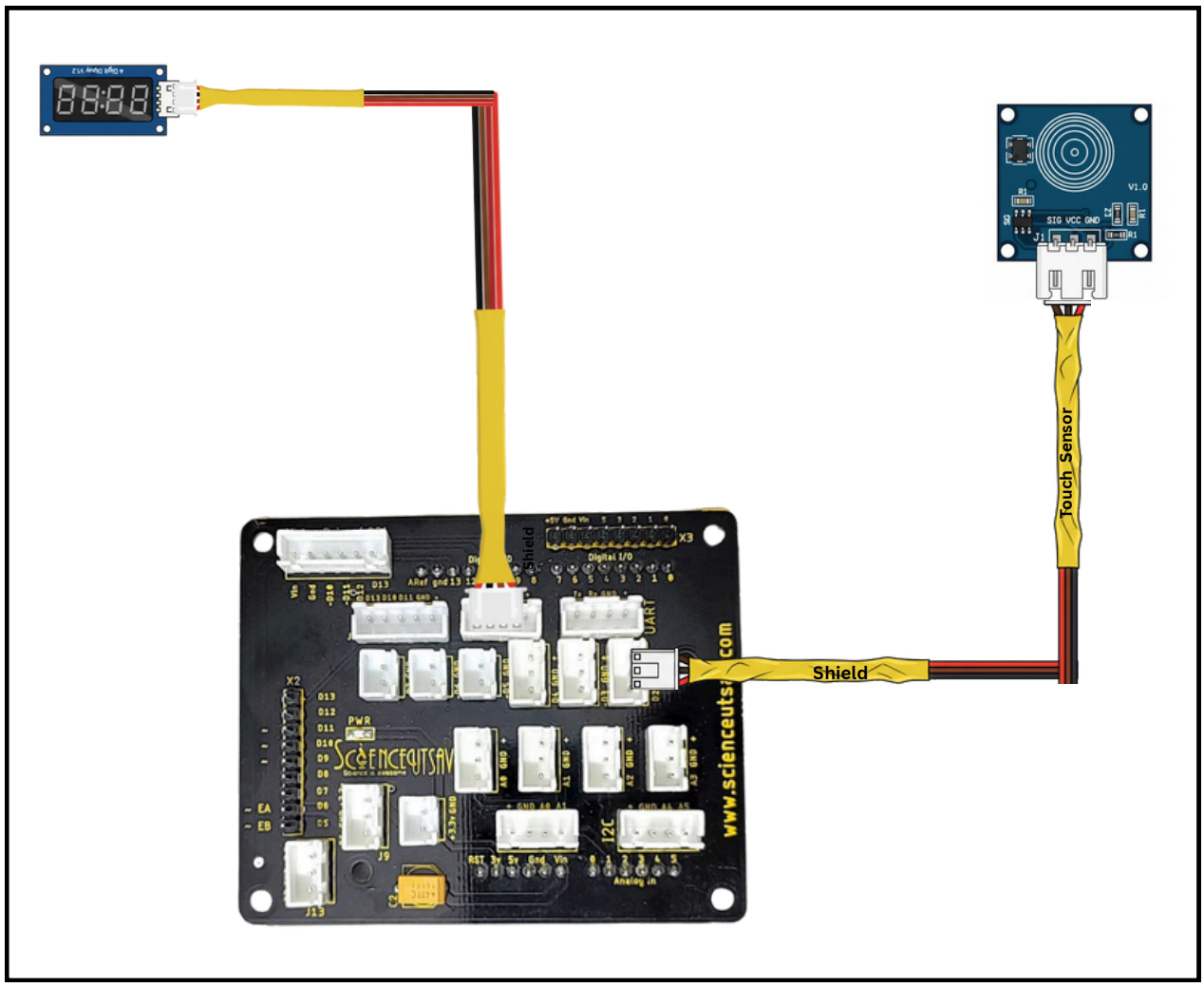
Code

```
// Session 7 Grade 6 Project 2
 // Speedometer
 // Car crosses IR1 then IR2 = speed value shown on 7 segment

 #include <TM1637Display.h>

 const int irSensor1 = 2;
 const int irSensor2 = 3;
 const int CLK = 9;
 const int DIO = 8;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite
 const float distanceCM = 20.0; // distance between two IR sensors

 TM1637Display display(CLK, DIO);

 unsigned long startTime = 0;
 bool timingStarted = false;
 bool lastIR1 = HIGH;
 bool lastIR2 = HIGH;

 void setup() {
 pinMode(irSensor1, INPUT);
 pinMode(irSensor2, INPUT);
 display.setBrightness(7);
 display.showNumberDec(0, true);
 }
```

```
void loop() {
 bool currentIR1 = digitalRead(irSensor1);
 bool currentIR2 = digitalRead(irSensor2);

 if (currentIR1 == IR_DETECTED && lastIR1 != IR_DETECTED) {
   startTime = millis();
   timingStarted = true;
 }

 if (timingStarted && currentIR2 == IR_DETECTED && lastIR2 != IR_DETECTED)
{
   unsigned long timeTaken = millis() - startTime;
   if (timeTaken > 0) {
     int speedValue = (distanceCM * 1000) / timeTaken; // simple cm/sec value
     if (speedValue > 9999) speedValue = 9999;
     display.showNumberDec(speedValue, true);
   }
   timingStarted = false;
   delay(500);
 }

 lastIR1 = currentIR1;
 lastIR2 = currentIR2;
 }
```

BEGINNER PROJECT

The circuit diagram of the project is given below

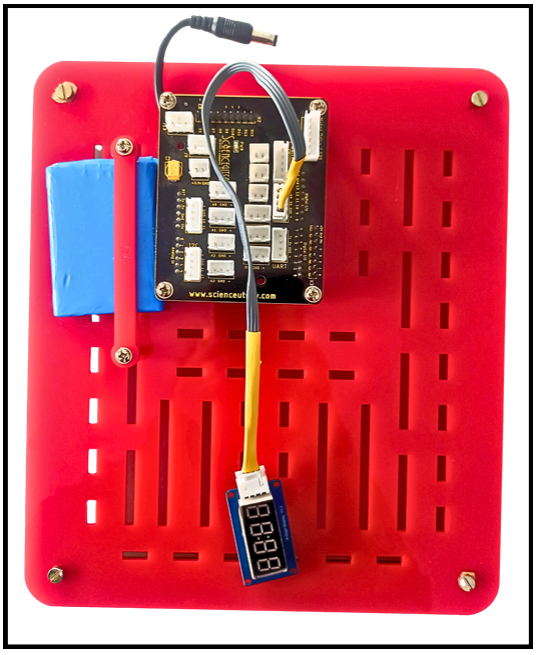


CONNECTION:

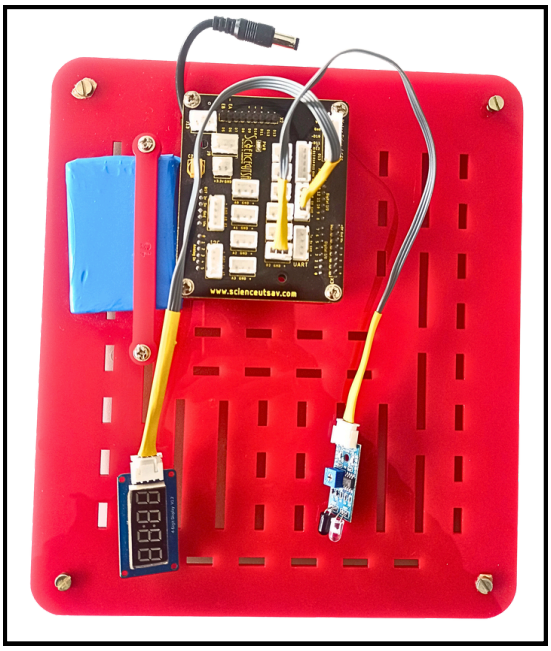
1.7 SEGMENT DISPLAY - D9 D8

2.Touch SENSOR- D2

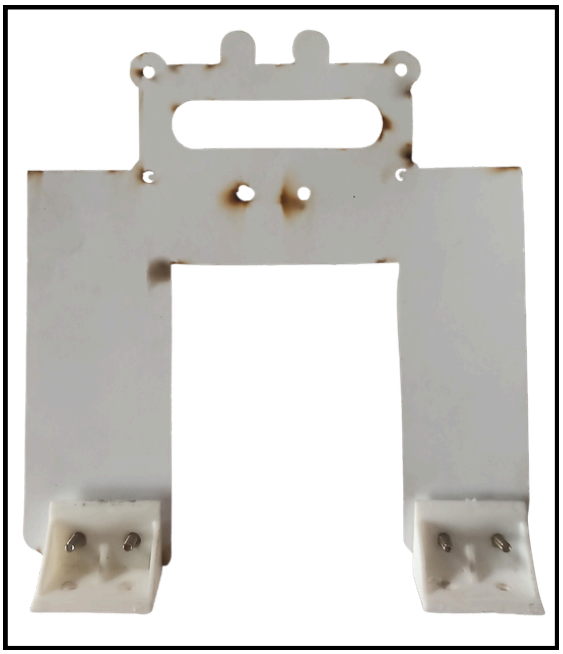
Step 1- Attach the 7 segment display to the aurdino as shown



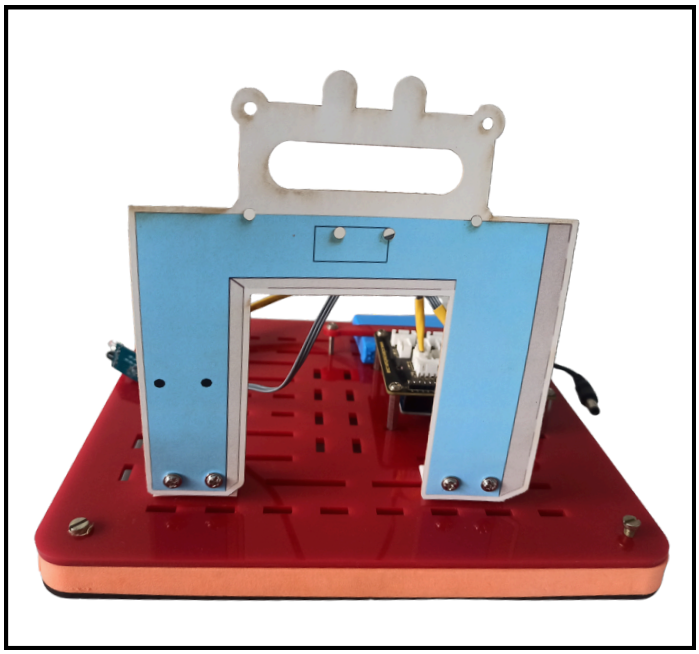
Step 2- Connect the IR sensor as well



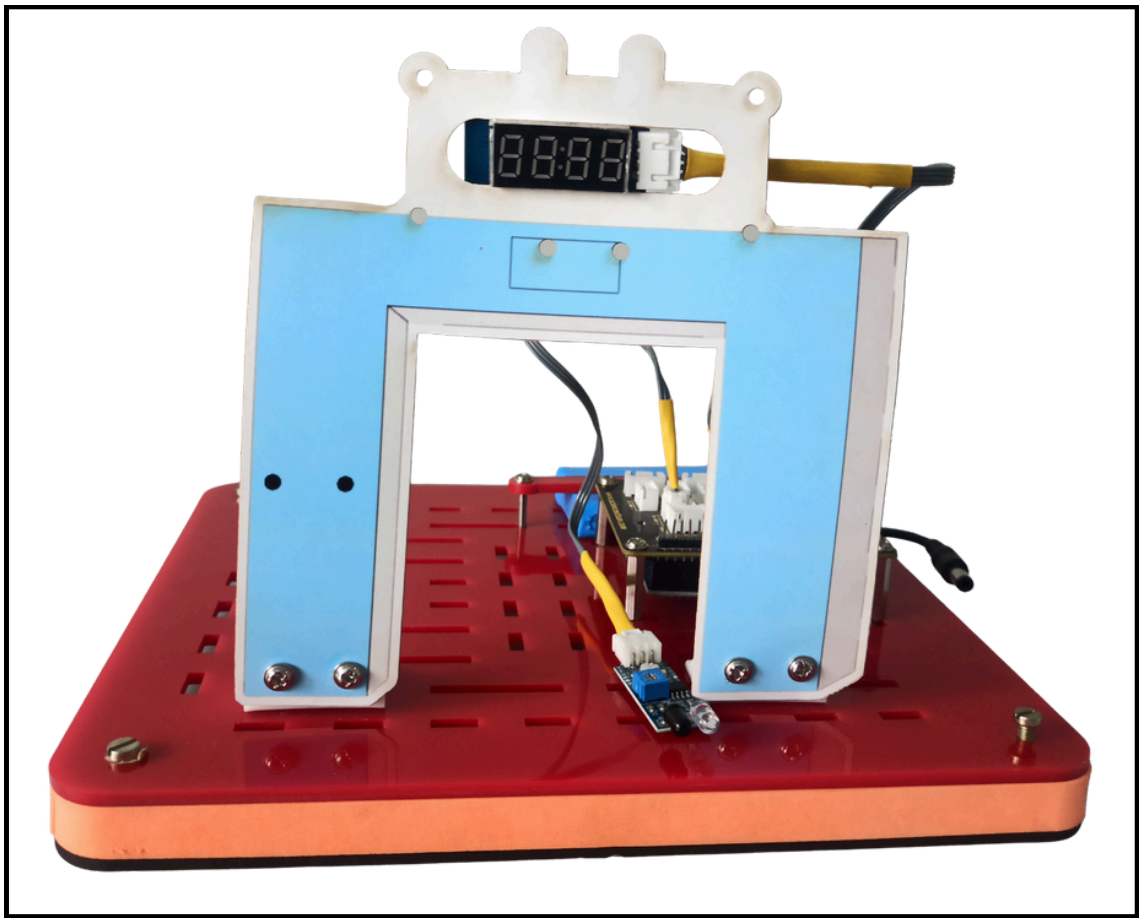
Step 3- Attach 2 L-clamps to each side of the template



Step 4- Fix the template to the base



Step 5- Place the display and sensor as shown. Upload the code and observe

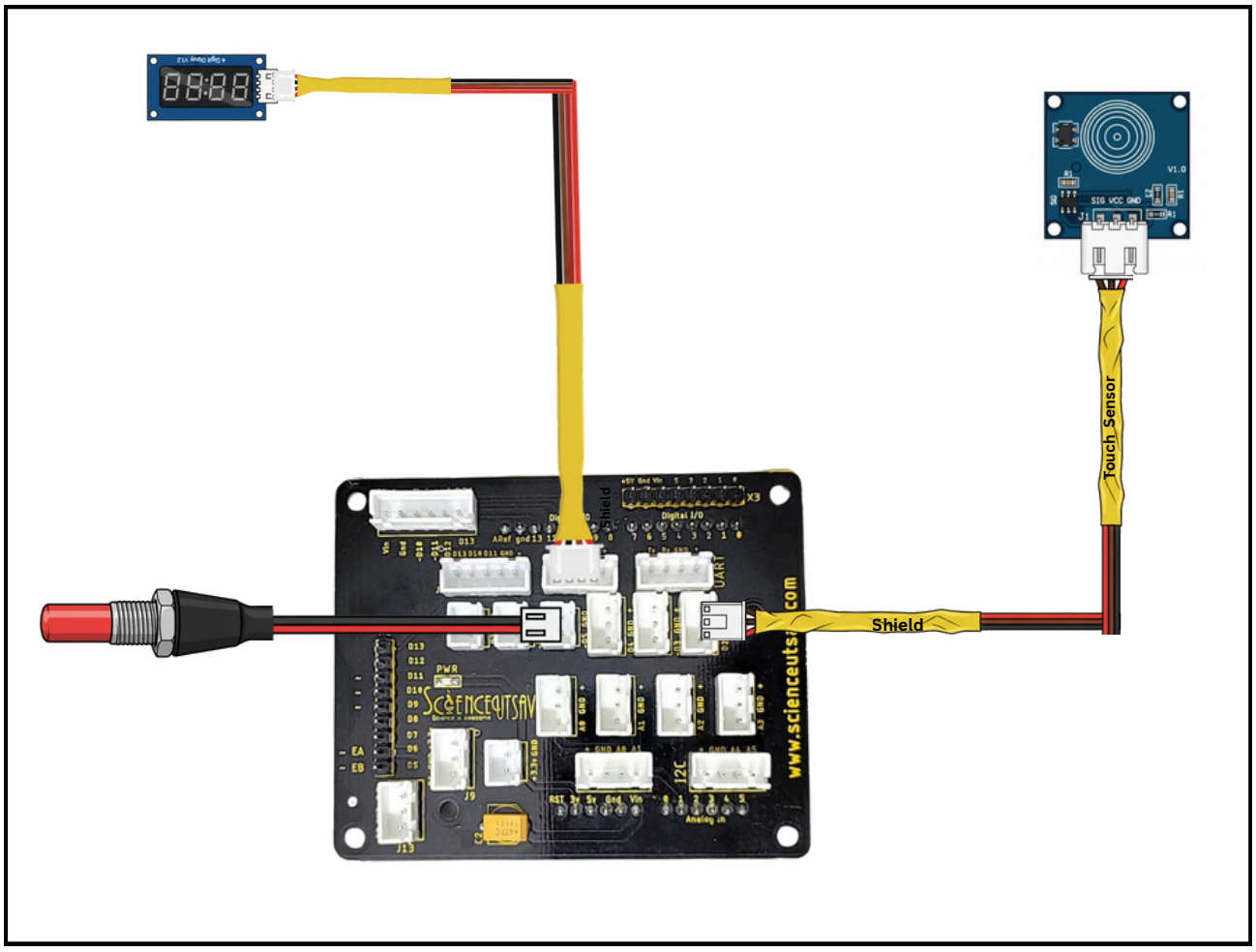


Code

```
// Session 7 Grade 7 Project 1
 // Arch Person Counter
 // Each person entering = count increases
 #include <TM1637Display.h>

 const int irSensor = 2;
 const int CLK = 9;
 const int DIO = 8;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 int personCount = 0;
 bool lastIR = HIGH;

 void setup() {
 pinMode(irSensor, INPUT);
 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 bool currentIR = digitalRead(irSensor);

 if (currentIR == IR_DETECTED && lastIR != IR_DETECTED) {
   personCount++;
   if (personCount > 9999) personCount = 0;
   display.showNumberDec(personCount, true);
   delay(300);
 }

 lastIR = currentIR;
 }
```

INTERMEDIATE PROJECT

The circuit diagram of the project is given below



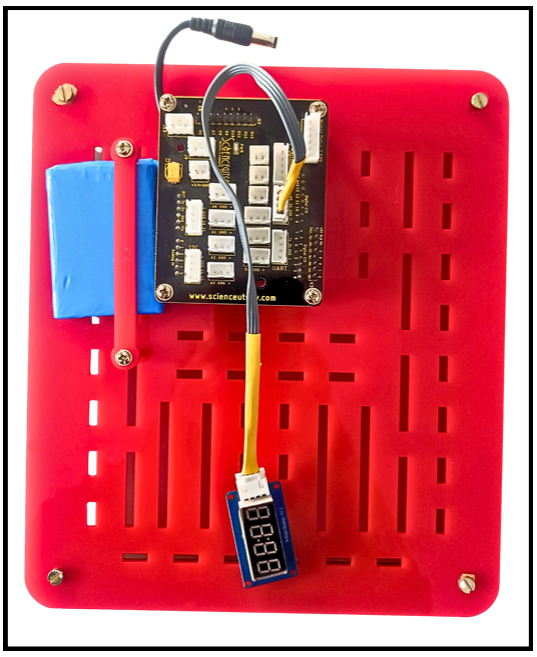
CONNECTION:

1.7 SEGMENT DISPLAY - D9 D8

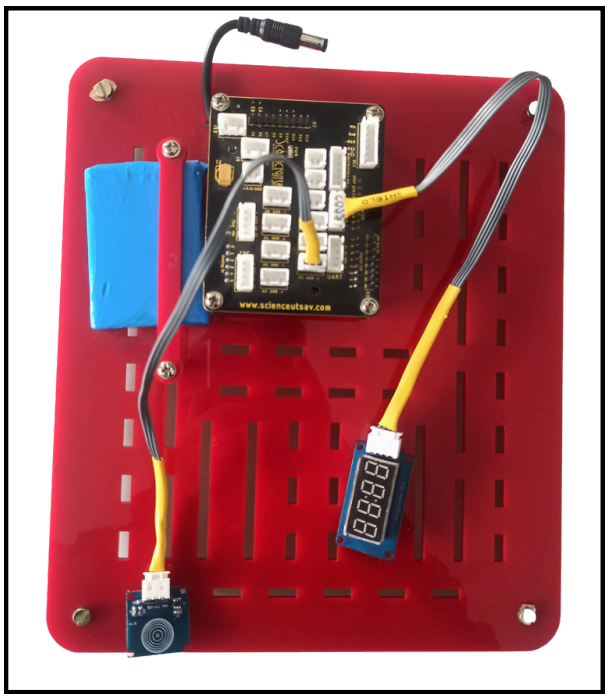
2.TOUCH SENSOR- D2

3.PUSH BUTTON IN D5

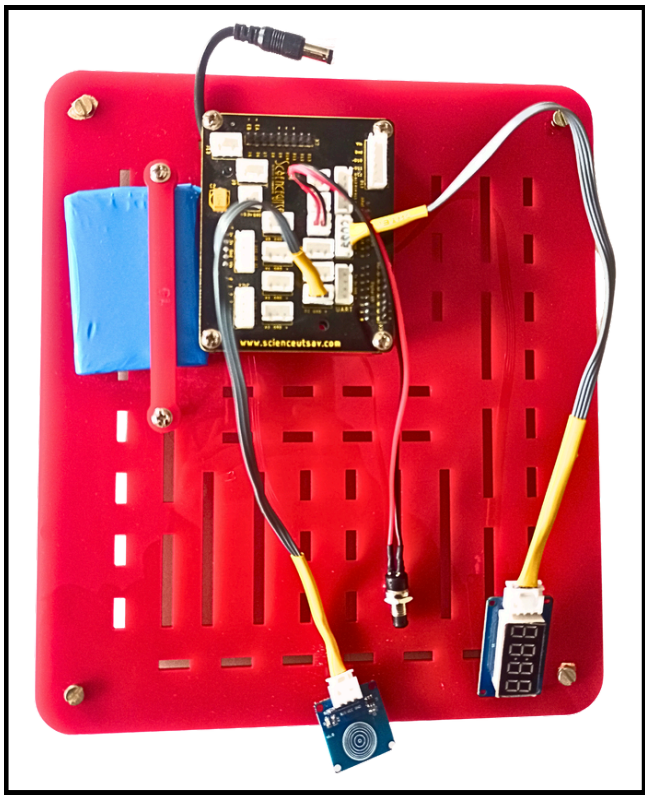
Step 1- Attach the 7 segment display to the aurdino as shown



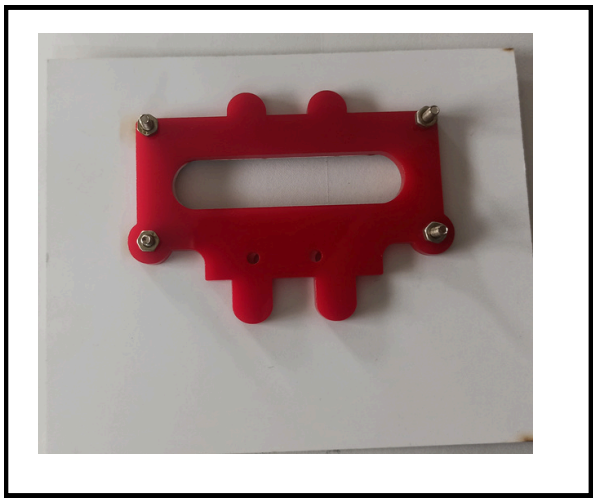
Step 2- Connect the touch sensor to the aurdino



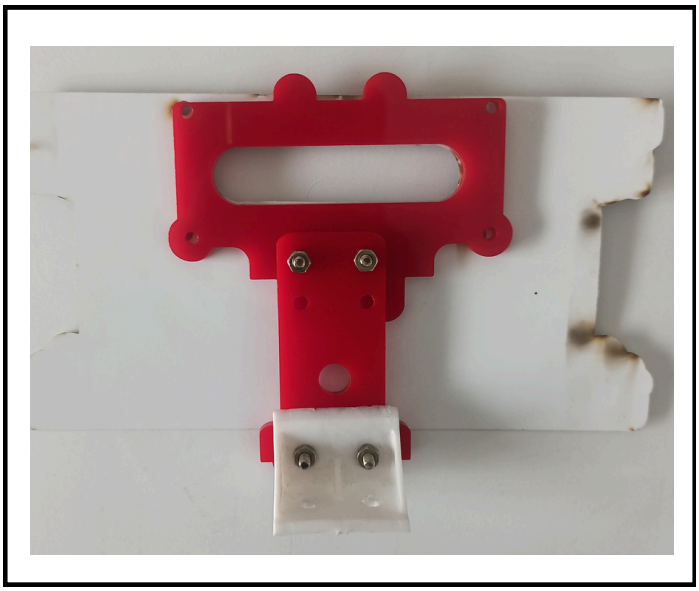
Step 3- Attach the push button as well



Step 4- Screw in the display stand to the template as shown



Step 5- Attach the stand shown below to the previous step



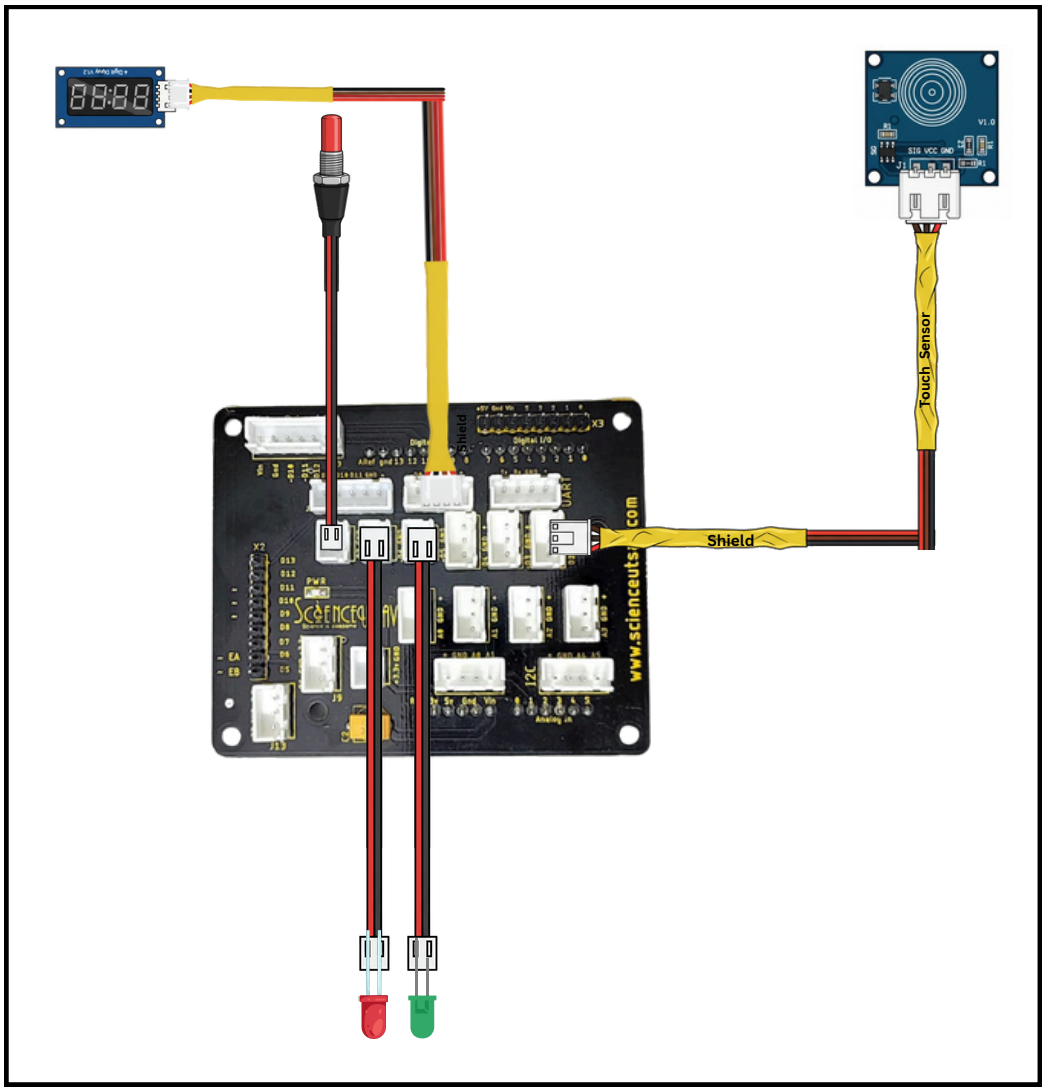
Step 6- Attach the template with the stand to the base and place the sensor, push button and display as shown. Upload the code and observe

Code

```
// Session 7 Grade 7 Project 2
 // Petrol Bunk Meter
 // Hold touch sensor = petrol value increases on 7 segment

 #include <TM1637Display.h>

 const int touchSensor = 2;
 const int CLK = 9;
 const int DIO = 8;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 TM1637Display display(CLK, DIO);

 int petrolValue = 0;
 unsigned long lastUpdate = 0;

 void setup() {
 pinMode(touchSensor, INPUT);
 display.setBrightness(7);
 display.showNumberDec(0, true);
 }

 void loop() {
 if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
   if (millis() - lastUpdate > 300) {
     petrolValue++;
     if (petrolValue > 9999) petrolValue = 0;
     display.showNumberDec(petrolValue, true);
     lastUpdate = millis();
   }
 }
 }
```

ADVANCE PROJECT

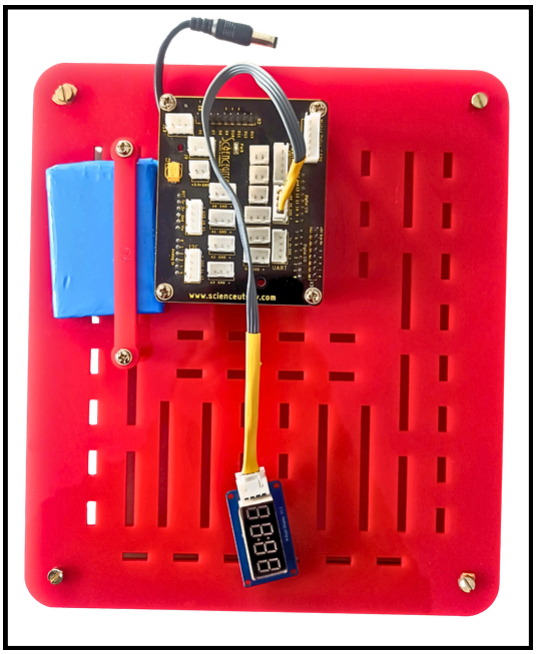
The circuit diagram of the project is given below



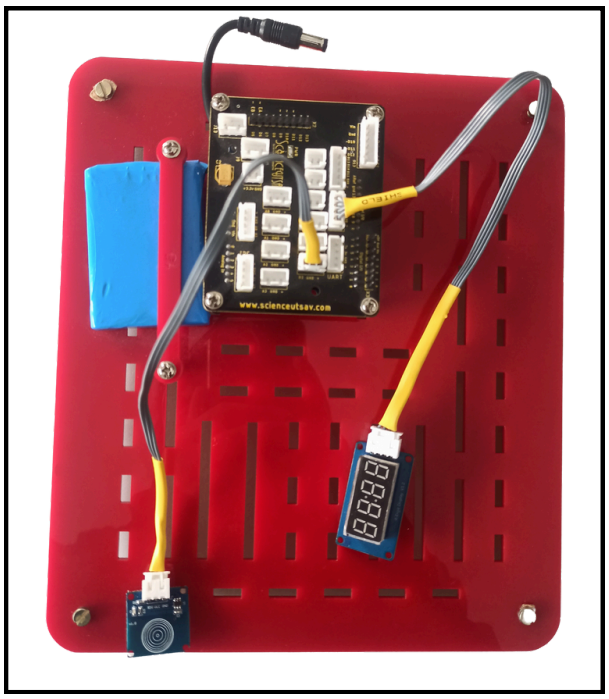
CONNECTION:

1. 7 SEGMENT DISPLAY - D9 D8
2. TOUCH SENSOR- D2
3. PUSH BUTTON - D7
4. RED DIP LED- D6
5. GREEN DIP LED- D5

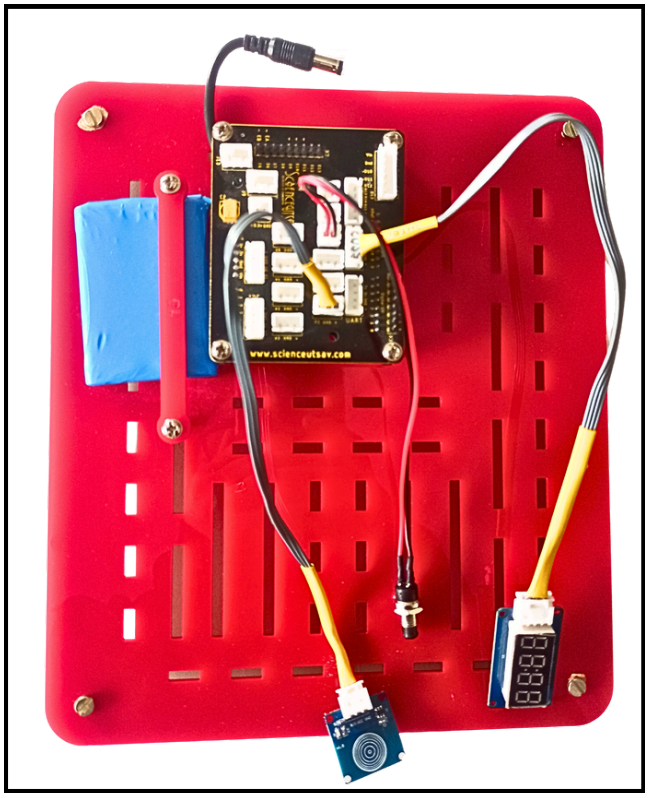
Step 1- Attach the 7 segment display to the aurdino as shown



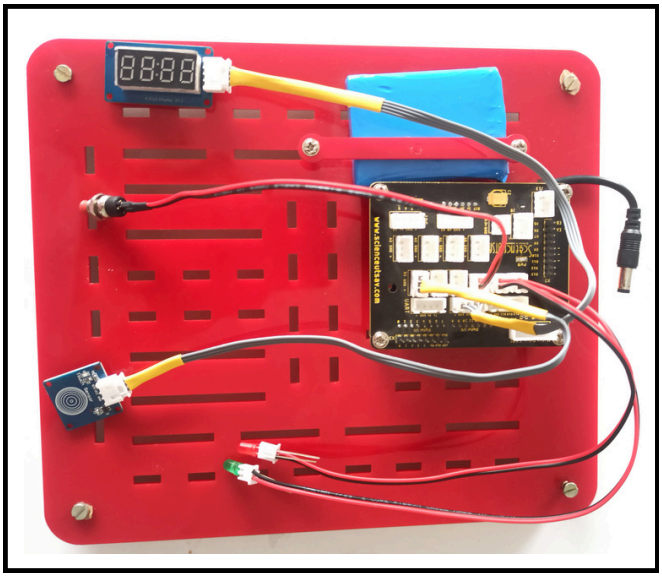
Step 2- Connect the touch sensor to the aurdino



Step 3- Attach the push button as well



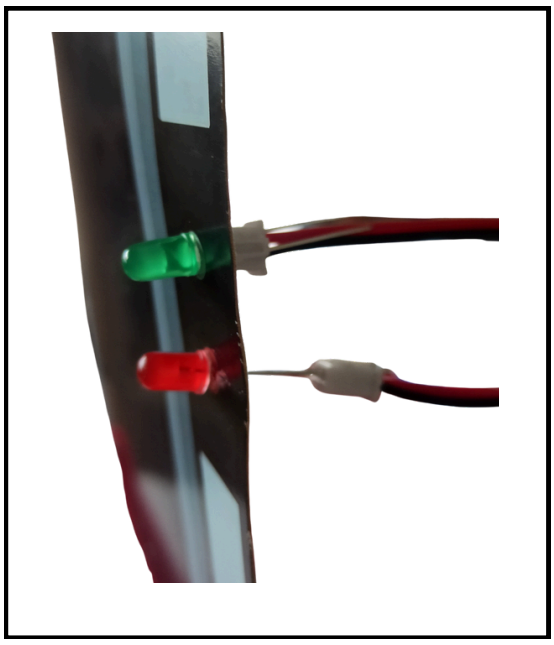
Step 4- Connect the red and green leds to the aurdino



Step 5- Screw in the display stand to the template as shown

Step 6- Attach the template with the stand to the base using a L-clamp

Step 7- Connect the dip LEDs to the template as shown



Step 8- Place the display, sensor and push button as shown

Step 9- Add a stand for the push button. Upload the code and observe

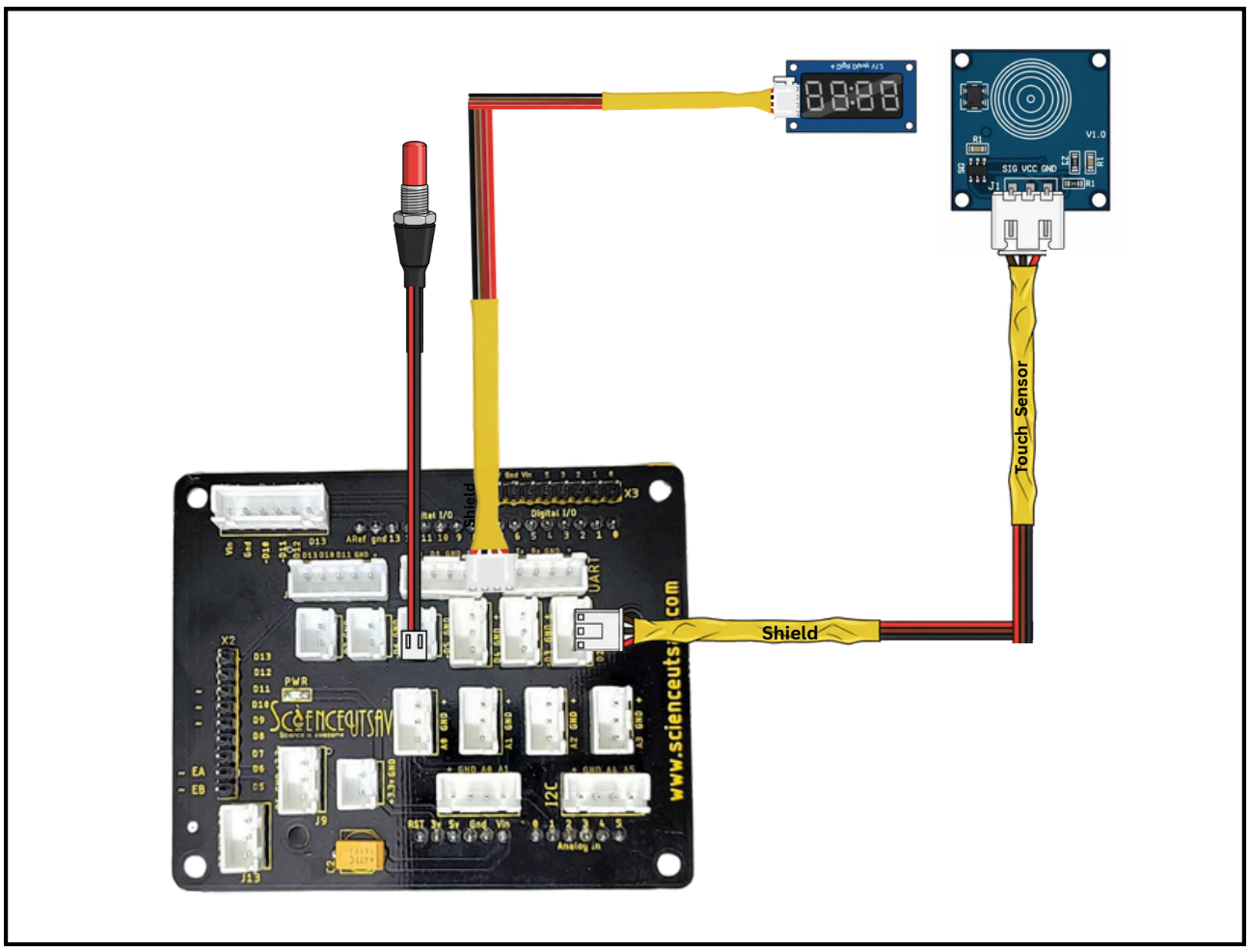
Code

```
// Session 7 Grade 8 Project 1
 // Elevator Floor System
 // Touch selects target floor, push confirms travel
 // Green LED = going up, Red LED = going down

 #include <TM1637Display.h>

 const int touchSensor = 2;
 const int pushButton = 7;
 const int redLED = 10;
 const int greenLED = 11;
 const int CLK = 9;
 const int DIO = 8;

 const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

 TM1637Display display(CLK, DIO);

 int currentFloor = 1;
 int targetFloor = 1;
 bool lastTouch = LOW;

 void setup() {
 pinMode(touchSensor, INPUT);
 pinMode(pushButton, INPUT_PULLUP);
 pinMode(redLED, OUTPUT);
 pinMode(greenLED, OUTPUT);

 display.setBrightness(7);
 display.showNumberDec(currentFloor, true);
 }
 void loop() {
 bool touchNow = digitalRead(touchSensor);
 if (touchNow == TOUCH_ACTIVE && lastTouch != TOUCH_ACTIVE) {
 targetFloor++;
```

```
if (targetFloor > 9) targetFloor = 1;
display.showNumberDec(targetFloor, true);
 delay(300);
}

if (digitalRead(pushButton) == LOW) {
moveToTargetFloor();
}

lastTouch = touchNow;
}

void moveToTargetFloor() {
if (targetFloor > currentFloor) {
digitalWrite(greenLED, HIGH);
 digitalWrite(redLED, LOW);

 while (currentFloor < targetFloor) {
currentFloor++;
display.showNumberDec(currentFloor, true);
delay(800);
 }
} else if (targetFloor < currentFloor) {
digitalWrite(greenLED, LOW);
 digitalWrite(redLED, HIGH);

while (currentFloor > targetFloor) {
currentFloor--;
display.showNumberDec(currentFloor, true);
  delay(800);
 }
}

digitalWrite(greenLED, LOW);
digitalWrite(redLED, LOW);
delay(300);
}
```

ADVANCE PROJECT

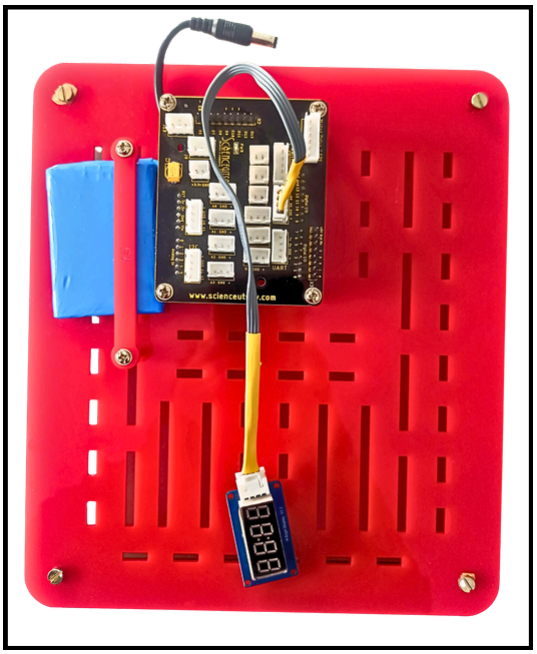
The circuit diagram of the project is given below



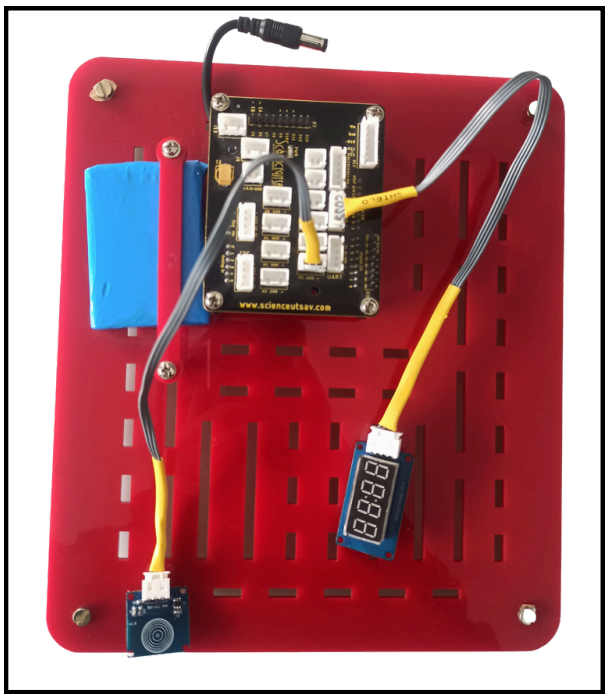
CONNECTION:

1. 7 SEGMENT DISPLAY - D9 D8
2. TOUCH SENSOR- D2
3. PUSH BUTTON - D7

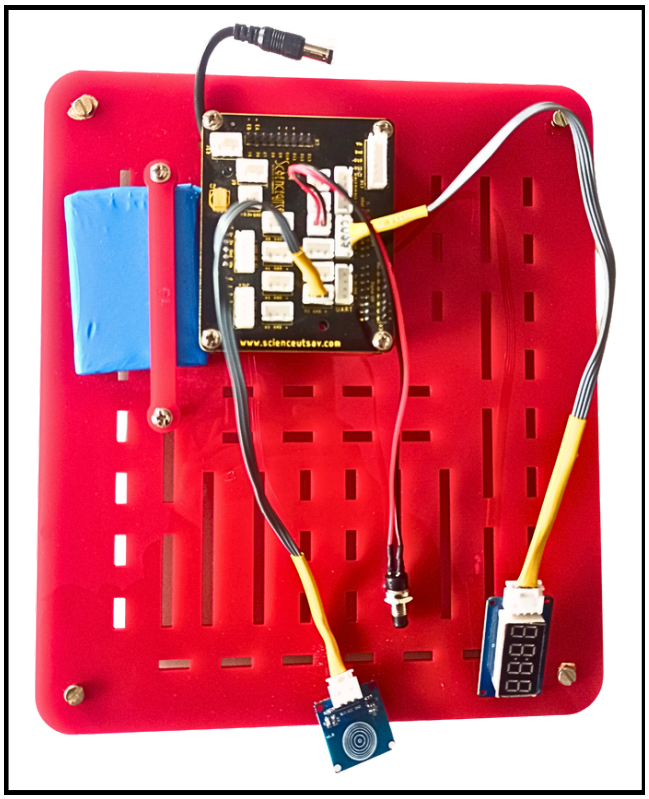
Step 1- Attach the 7 segment display to the aurdino as shown



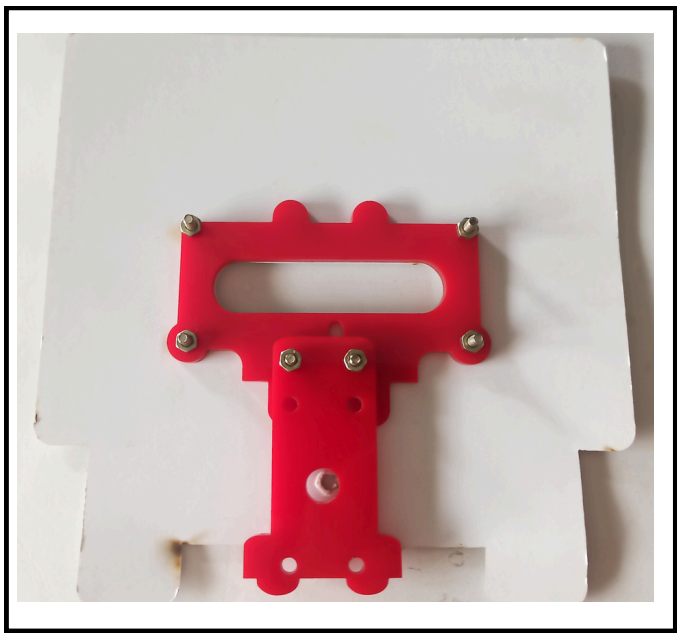
Step 2- Connect the touch sensor to the aurdino



Step 3- Attach the push button as well

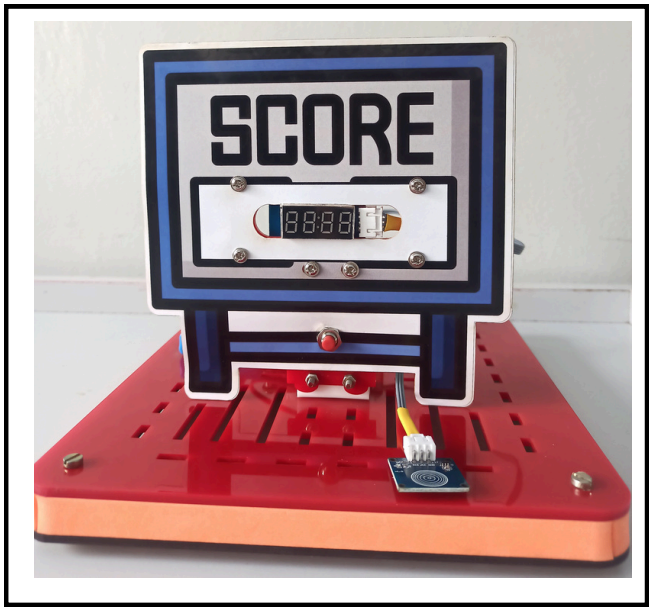


Step 4- Screw in the display stand to the template as shown



Step 5- Attach the template to the base using a L-clamp

Step 6- Place the display, button and touch sensor as shown below



Step 7- Upload the code and observe

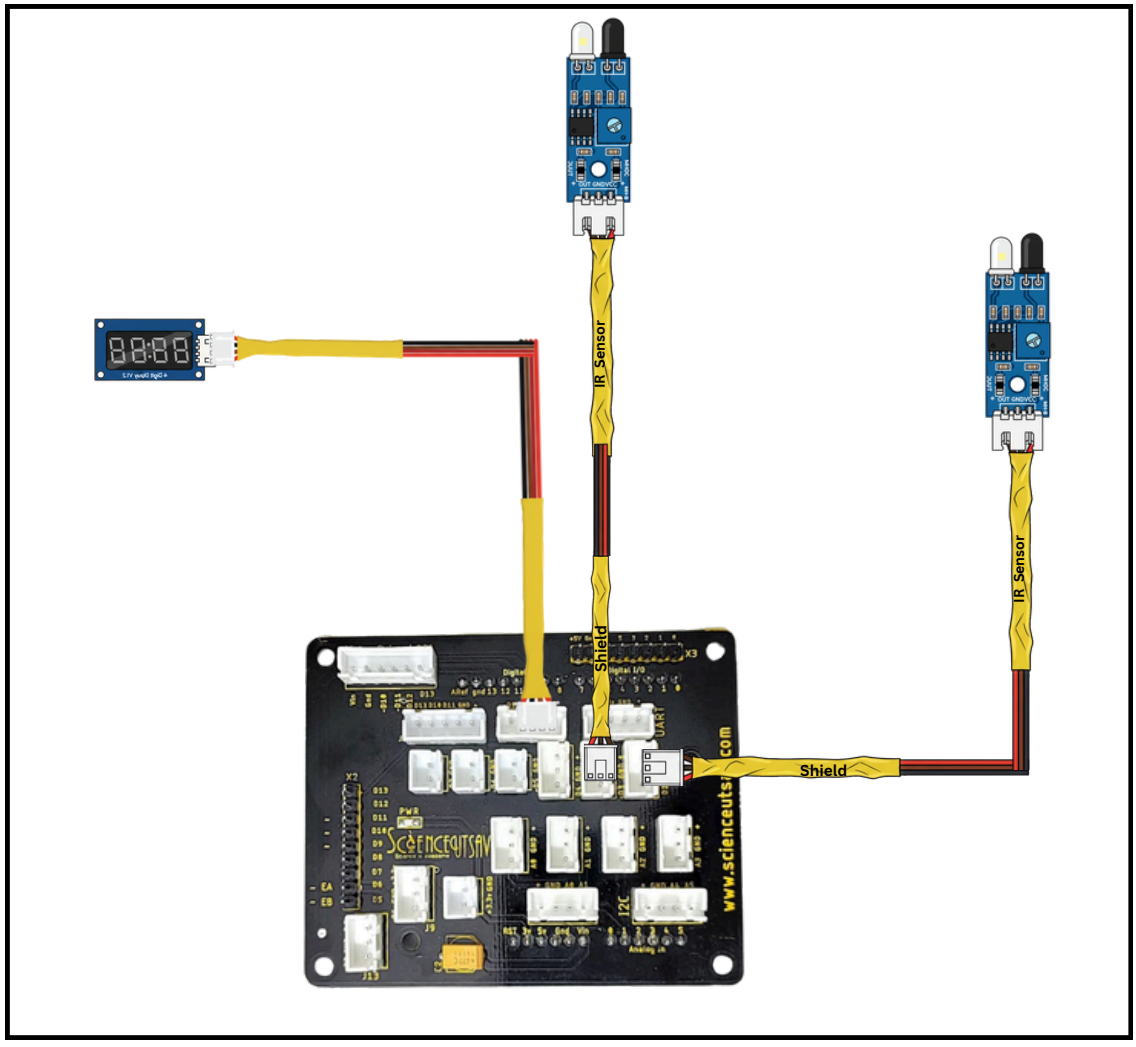
Code

```
// Session 7 Grade 8 Project 2
// Scoreboard
// Touch selects Team A or Team B, Push increases selected team score
// Display format: A score on left two digits, B score on right two digits

#include <TM1637Display.h>

const int touchSensor = 2;
const int pushButton = 5;
const int CLK = 9;
const int DIO = 8;

const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

TM1637Display display(CLK, DIO);

int scoreA = 0;
int scoreB = 0;
int selectedTeam = 1; // 1 = Team A, 2 = Team B
bool lastTouch = LOW;
bool lastButton = HIGH;

void setup() {
pinMode(touchSensor, INPUT);
pinMode(pushButton, INPUT_PULLUP);
display.setBrightness(7);
showScore();
}

void loop() {
bool touchNow = digitalRead(touchSensor);
bool buttonNow = digitalRead(pushButton);
```

```
if (touchNow == TOUCH_ACTIVE && lastTouch != TOUCH_ACTIVE) {
  selectedTeam = (selectedTeam == 1) ? 2 : 1;
  blinkSelectedTeam();
  delay(300);
}

if (buttonNow == LOW && lastButton == HIGH) {
  if (selectedTeam == 1) {
    scoreA++;
    if (scoreA > 99) scoreA = 0;
  } else {
    scoreB++;
    if (scoreB > 99) scoreB = 0;
  }
  showScore();
  delay(300);
}

lastTouch = touchNow;
lastButton = buttonNow;
}

void showScore() {
int displayValue = scoreA * 100 + scoreB;
display.showNumberDec(displayValue, true);
}

void blinkSelectedTeam() {
display.showNumberDec(selectedTeam, true);
delay(500);
showScore();
}
```

EXPERT PROJECT

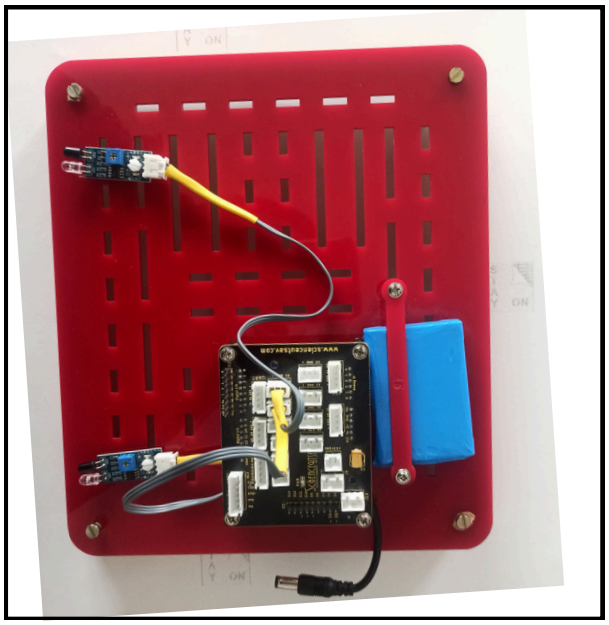
The circuit diagram of the project is given below



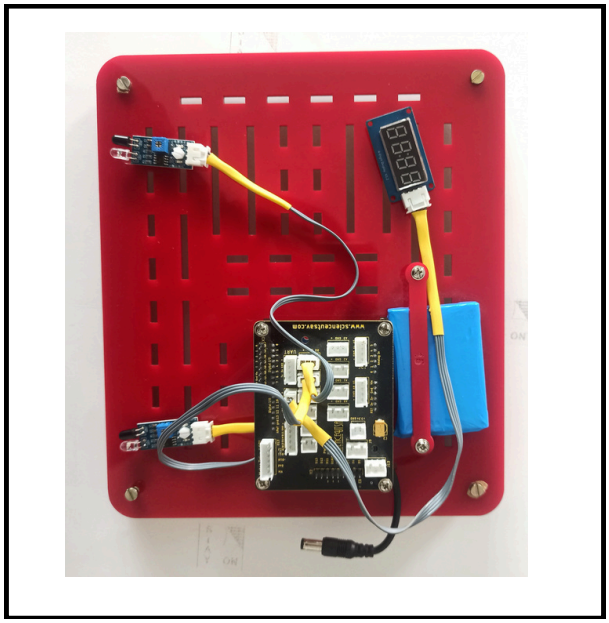
CONNECTION:

1. IR sensor- D2, D3
2. 7-segment display- D8, D9

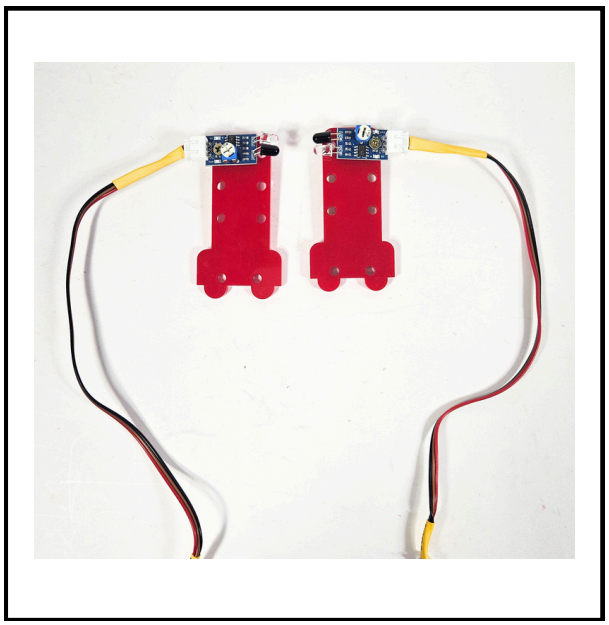
Step 1- Connect the 2 IR sensors to the shield



Step 2- Connect the 7-segment display as shown

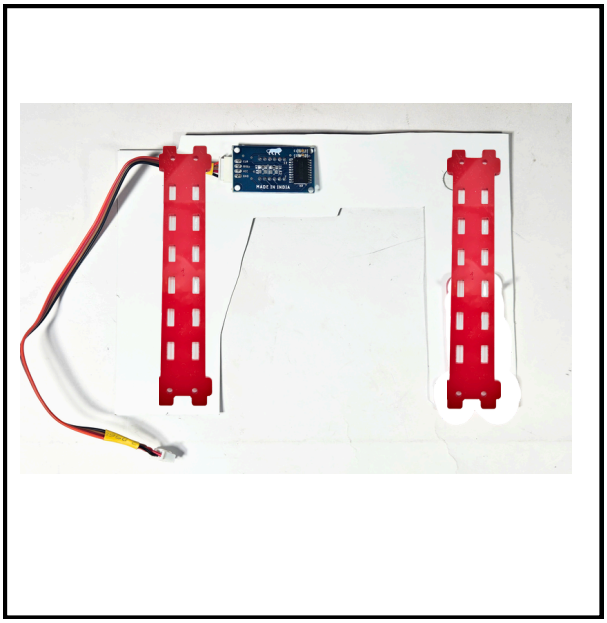


Step 3- Attach the sensors to the stands as shown

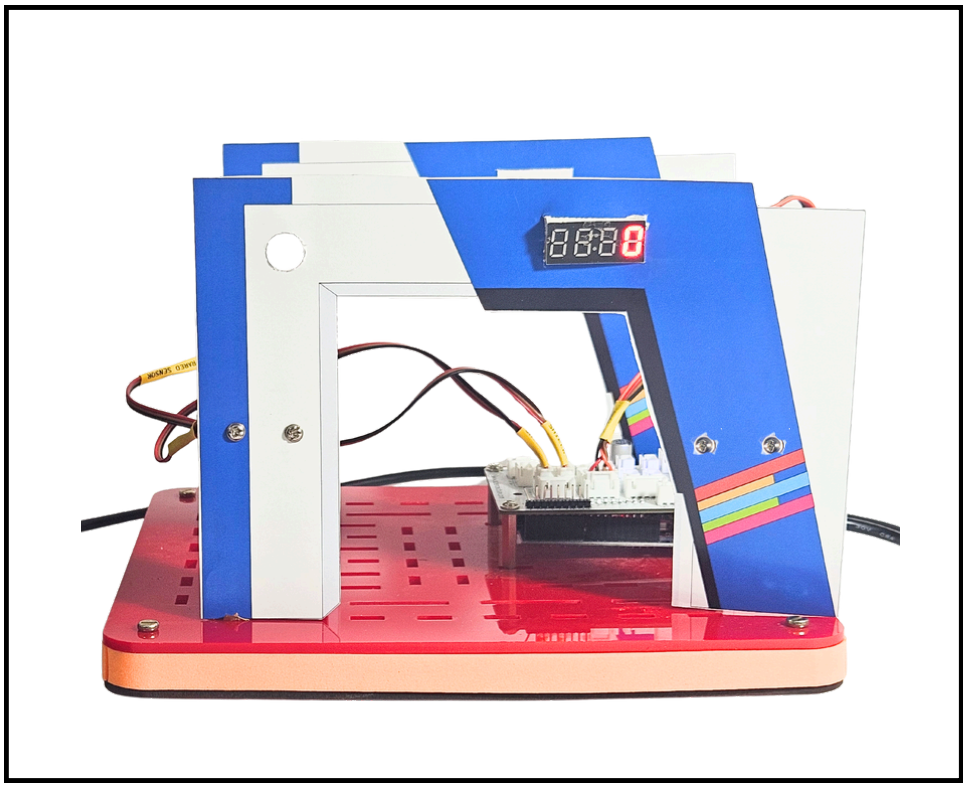


Step 4- Fix the stands to the board on opposite sides

Step 5- Fix the stands to both the template and attach the display



Step 6- Fix the templates to the board as shown



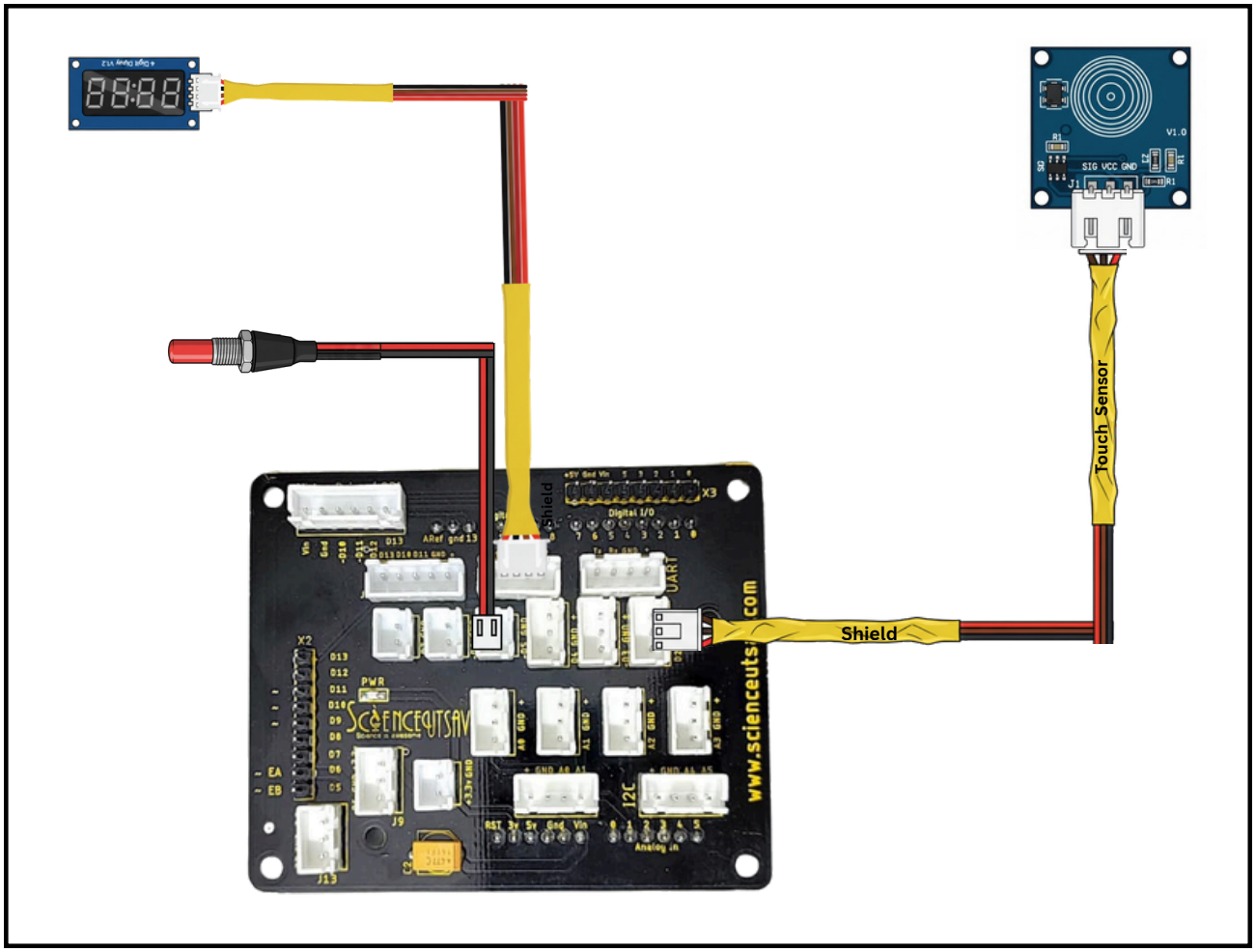
Code

```
// Session 7 Grade 9 Project 1
 // Entry Exit Occupancy Counter
 // IR1 = person entered, IR2 = person left
 // Display shows number of persons inside

 #include <TM1637Display.h>

 const int entryIR = 2;
 const int exitIR = 3;
 const int CLK = 9;
 const int DIO = 8;

 const int IR_DETECTED = LOW;  // change to HIGH if opposite

 TM1637Display display(CLK, DIO);

 int insideCount = 0;
 bool lastEntry = HIGH;
 bool lastExit = HIGH;

 void setup() {
 pinMode(entryIR, INPUT);
 pinMode(exitIR, INPUT);
 display.setBrightness(7);
 display.showNumberDec(insideCount, true);
 }
```

```
void loop() {
 bool entryNow = digitalRead(entryIR);
 bool exitNow = digitalRead(exitIR);

 if (entryNow == IR_DETECTED && lastEntry != IR_DETECTED) {
 insideCount++;
 if (insideCount > 9999) insideCount = 9999;
 display.showNumberDec(insideCount, true);
 delay(300);
 }

 if (exitNow == IR_DETECTED && lastExit != IR_DETECTED) {
 if (insideCount > 0) insideCount--;
  display.showNumberDec(insideCount, true);
 delay(300);
 }

 lastEntry = entryNow;
 lastExit = exitNow;
 }
```

EXPERT PROJECT

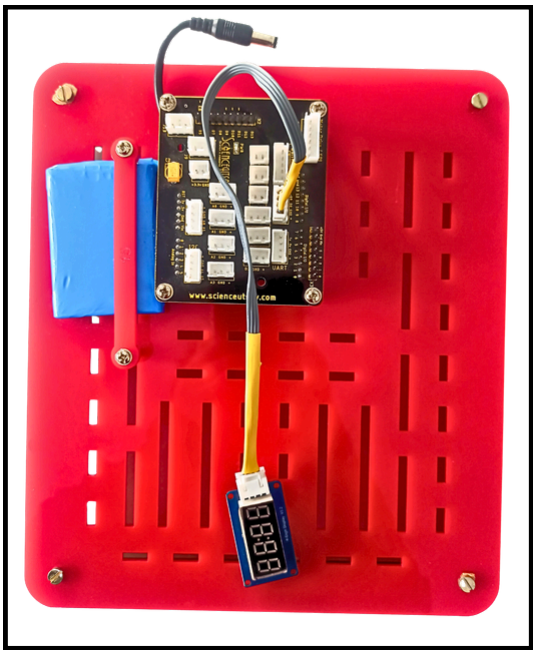
The circuit diagram of the project is given below



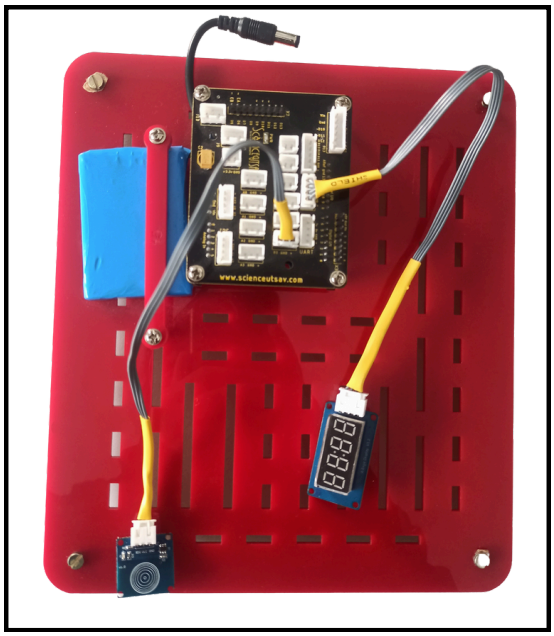
CONNECTION:

1. 7 SEGMENT DISPLAY - D9 D8
2. TOUCH SENSOR- D2
3. PUSH BUTTON IN D5

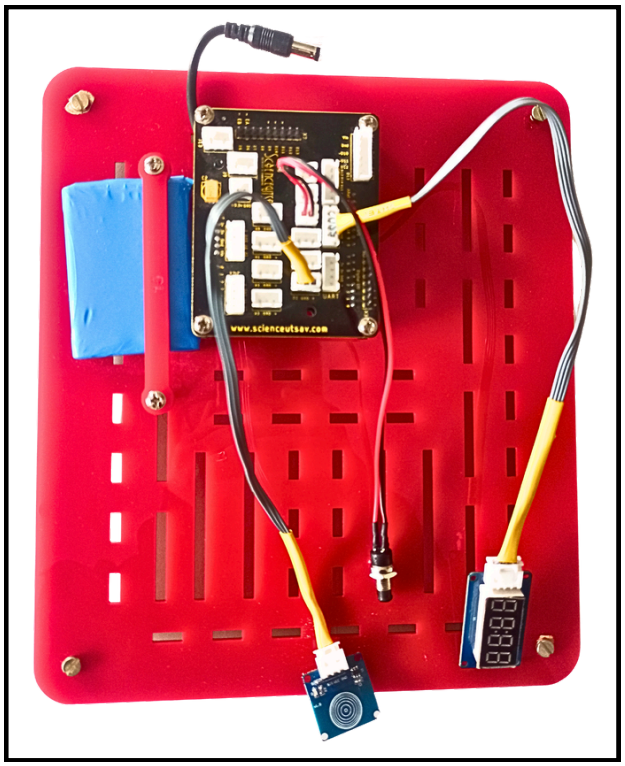
Step 1- Attach the 7 segment display to the aurdino as shown



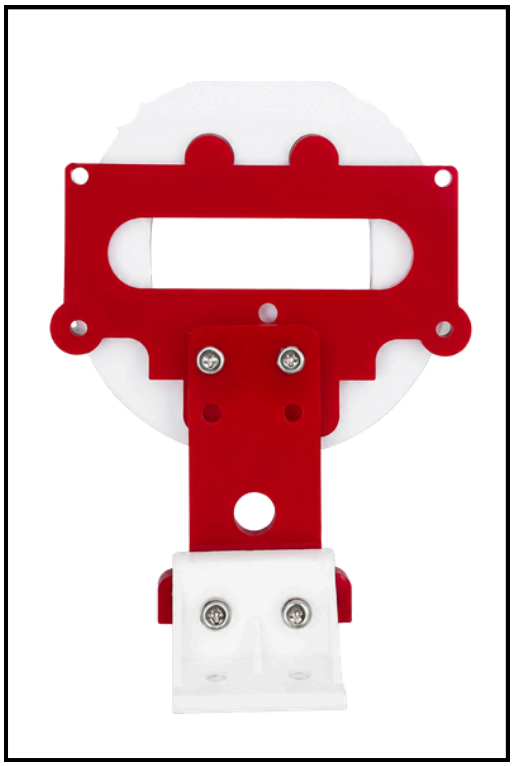
Step 2- Connect the touch sensor to the aurdino



Step 3- Attach the push button as well



Step 4- Screw in the display stand to the template as shown



Step 5- Connect the template stand to the base

Step 6- Upload the code and observe

Code

```
// Session 7 Grade 9 Project 2
// Stopwatch
// Push button cycles: START -> PAUSE -> RESET
// Touch sensor can also reset anytime

#include <TM1637Display.h>

const int touchSensor = 2;
const int pushButton = 5;
const int CLK = 9;
const int DIO = 8;

const int TOUCH_ACTIVE = HIGH; // change to LOW if opposite

TM1637Display display(CLK, DIO);

int mode = 0; // 0 reset, 1 running, 2 paused
unsigned long startMillis = 0;
unsigned long pausedTime = 0;
unsigned long elapsedMillis = 0;
bool lastButton = HIGH;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(pushButton, INPUT_PULLUP);
  display.setBrightness(7);
  showTime(0);
}
```

```
void loop() {
  bool buttonNow = digitalRead(pushButton);

  if (buttonNow == LOW && lastButton == HIGH) {
    if (mode == 0) {
      mode = 1;
      startMillis = millis();
      elapsedMillis = 0;
    } else if (mode == 1) {
      mode = 2;
      pausedTime = elapsedMillis;
    } else {
      mode = 0;
      elapsedMillis = 0;
      showTime(0);
    }
    delay(300);
  }

  if (digitalRead(touchSensor) == TOUCH_ACTIVE) {
    mode = 0;
    elapsedMillis = 0;
    showTime(0);
    delay(300);
  }

  if (mode == 1) {
    elapsedMillis = pausedTime + (millis() - startMillis);
    showTime(elapsedMillis / 1000);
  }

  lastButton = buttonNow;
}

void showTime(unsigned long totalSeconds) {
  int minutes = (totalSeconds / 60) % 100;
  int seconds = totalSeconds % 60;
  int value = minutes * 100 + seconds;
  display.showNumberDecEx(value, 0b01000000, true);
}
```